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Experiment 6: Mobile Network Handoff Simulation Using Figma

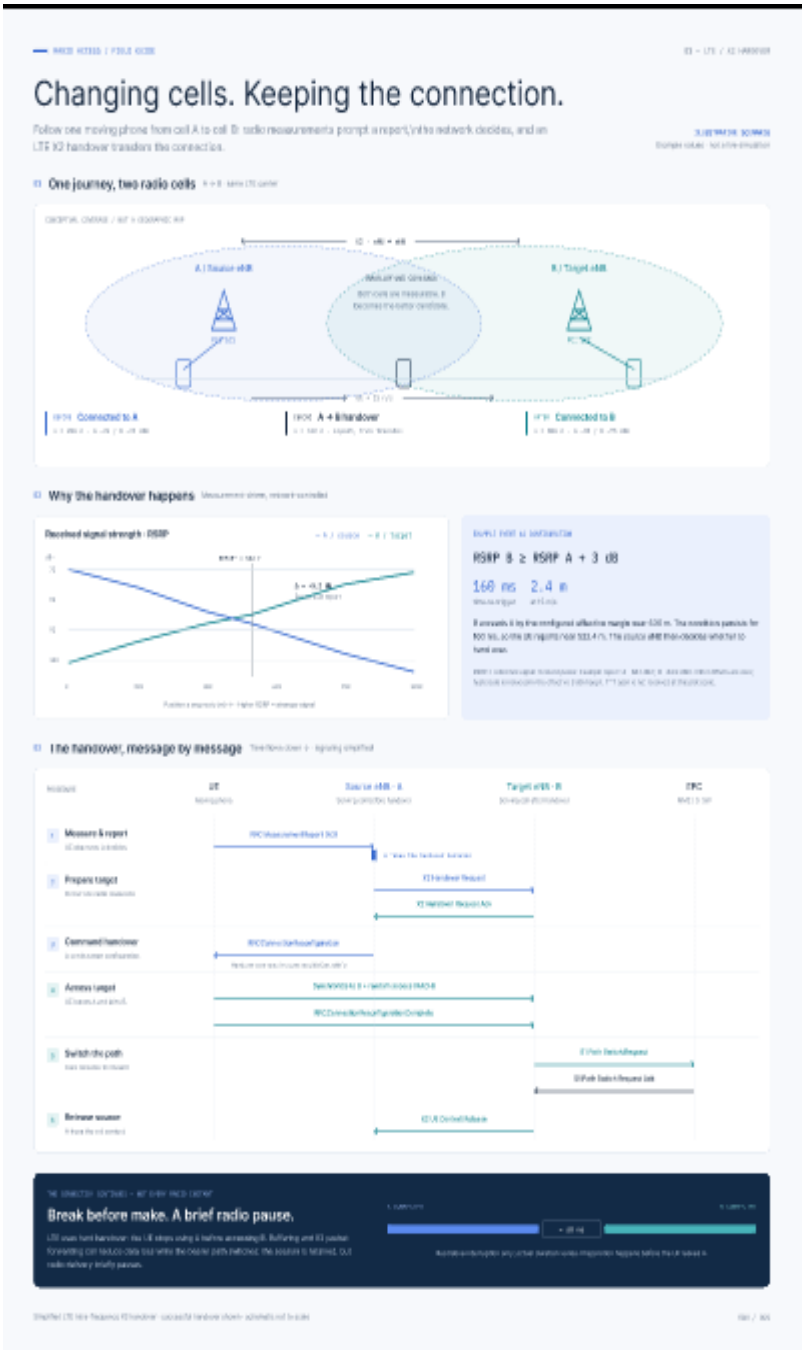
Aim:

To design and simulate a handoff scenario in a mobile communication network using Figma and visualize the process of transferring a mobile device's connection from one base station to another.

Procedure:

1. Open Figma and create a new design file with a desktop-sized frame and a light background.
2. Create a title section named **"Changing Cells, Keeping the Connection"** with a short description of mobile network handoff.
3. Draw two overlapping cell coverage areas representing **Base Station A** and **Base Station B**.
4. Add icons or shapes for the mobile device and base stations, and connect them using lines to illustrate wireless communication.
5. Create a signal-strength graph showing how the signal from Base Station A decreases while the signal from Base Station B increases as the mobile device moves.
6. Add a handoff decision panel displaying example signal-strength values and a threshold for initiating the handoff.
7. Design a message-sequence diagram showing the steps: measurement report, handoff preparation, handoff command, access to the target cell, switching the connection, and resource release.
8. Use different colors to distinguish the source cell, target cell, signal paths, and handoff messages.
9. Add a final status panel indicating that the mobile device has successfully connected to Base Station B without interrupting the ongoing communication.
10. Arrange all elements neatly, following the layout of the reference image, and export the completed design.

Output:



Result:

A mobile network handoff simulation dashboard was designed in Figma, illustrating cell coverage, signal-strength variation, handoff decision-making, message exchange, and the transfer of an active mobile connection between two base stations.